

# Neel Patel

neel@notneelpatel.xyz | notneelpatel.xyz/portfolio | github.com/NotNeelPatel

## Education

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**Carleton University** – B.Eng, High Distinction, Computer Systems Engineering, Co-op      Sept 2021 – Apr 2026

- Recipient of Entrance Scholarship, Dean's Honour List

## Experience

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**Research Assistant** | Carleton Immersive Media Studio – Ottawa, ON      May 2026 – Present

- Using Natural Language Processing (NLP) techniques with LLMs to assist with data collection

**Software Developer Student Co-op** | Solace – Ottawa, ON      Winter 2024, Fall 2024, Summer 2025

- Streamlined SVN to Git migration, cutting a multi-week process down to a single-day task
- Architected, implemented, and tested features for Solace C and Python APIs, optimizing bandwidth usage and enabling message compression
- Led the migration and open-source release of the Wireshark SMF Plugin, improving community initiatives
- Automated CI/CD pipelines and releases using GitHub Actions and Jenkins, enabling automated developer QA and plugin deployment

**Full-Stack Developer Co-op** | LearnExperts – Ottawa, ON      Summer 2024

- Automated testing and report generation with GitHub Actions, increasing bug visibility and reducing regressions
- Enhanced bug tracking by implementing AWS RUM with custom metadata for error logs
- Developed end-to-end Cypress tests for key interactive workflows, reducing manual test requirements by 40%

**Jr. Test Analyst Co-op** | Immigration, Refugees, and Citizenship Canada – Ottawa, ON      Summer 2023

- Ensured reliability of government application used by 10 000+ employees through verification of manual tests
- Collaborated with IT and development teams to formalize and document test procedures, improving traceability

**Jr. Software Developer, High School Co-op** | NorthStar Utilities – Ottawa, ON      Winter 2020

- Standardized customer .properties files by developing a Java utility that parses and cleans data
- Leveraged Git, Docker, and agile methodology, gaining early experience in developer practices

## Projects

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- **Wallpaper Theme Converter:** JavaScript website that changes the colour scheme of wallpapers, with 140+ stars on GitHub
- **AI Capstone Project:** Architected a multi-agent RAG system to analyze City of Ottawa's asset management documents
- **AI Raspberry Pi Bamboo Plant:** Engineered an IoT monitoring system with AI voice capabilities to monitor water levels and generate personalized voice alerts
- **Ross Video FPGA Challenge:** Designed a SystemVerilog and Python system to simulate GIF rendering via a custom ROM for the FPGA, winning team
- **FPGA Colour Filter:** Used Zybo Z7-10 + Verilog to develop a colour filter LUT system with HDMI output
- **Team Lead, School Projects:** Managed multiple engineering teams (Capstone, SYSC, ECOR courses) to achieve A+ project outcomes

## Skills

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**Languages:** Python, Java, C, C++, ARM Assembly, Verilog, BASH, HTML, CSS, JavaScript

**Project Management:** Git, SVN, Jira, Confluence

**Hardware/Embedded:** Vivado, Zybo FPGA, Raspberry Pi, Arduino

**Other Tools:** GitHub Actions, Jenkins, Roo/Cline/Copilot, RAG, ADK, LangChain, Wireshark, Testing Frameworks, GDB, Azure, Web Scraping